

ANBU MURUGESAN

Full-Stack Data Engineer · Analytics & AI
9172443599 · anbuja18@gmail.com · anbuja.github.io/portfolio

PROFILE

Full-stack data engineer with three years of experience delivering analytics, automation and reporting solutions for global banking. Currently on Citi's Global Liquidity Reporting System (GLRS) Data & Analytics team (via LTI), building end-to-end analytics platforms and owning all AI and automation initiatives across the reporting function. Sole driver of a production full-stack platform and sole driver of all AI-assisted development and workflow design tooling. Combines deep SQL engineering (Oracle, Trino/Starburst) with front-to-back product development (React + Angular + FastAPI) and hands-on regulatory domain knowledge across LCR, NSFR, ILST, RLAP, TLST and FR2052a.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (ES6+), TypeScript, HTML/CSS

Frontend: React, Angular, responsive design, component architecture, state management

Data & Query: Trino / Starburst, Oracle SQL, query optimization, CTE restructuring, predicate pushdown, federated cross-catalog queries, Parquet / Hive metastore

Full-Stack: FastAPI, REST API design, backend proxying, authentication & session handling, audit logging, rate limiting

BI & Reporting: Tableau (dashboard development, TDS / data-source analysis), automated commentary generation, data reconciliation tooling

AI & Automation: Process automation, Python scripting, AI-assisted development, prompt engineering, LLM-based code generation, AI workflow design

Platform: Helm, GitLab, Harness / Lightspeed CI-CD, Red Hat OpenShift / ECS, containerized deployment, Jira

Domain: LCR, NSFR, ILST, RLAP, TLST, ALCO, RLEN, FR2052a; HQLA, inflow/outflow and surplus computation, collateral-level haircuts, trapping logic

PROFESSIONAL EXPERIENCE

Data Engineer 2 | Citi (GLRS) · Vendor: LTI

July 2024 – Current · Promoted from Developer to Data Engineer 2 · Awarded Best Team Player (2024–25)

Flagship Project — GLRS Analytics Hub (React + FastAPI internal platform)

Sole developer of a full-stack analytics platform that replaced an ad-hoc prototype and consolidated eight separate reporting workflows into a single secure internal web application. Owned every layer — requirements, architecture, front end, backend, security and release.

- Architecture — React single-page front end backed by a FastAPI service layer, with offline-first design and backend proxy routing.
- Dynamic Dashboard & Forecasting — configurable dashboard builder with real-time liquidity metrics.
- Automated Commentary Engine — multi-sheet LCR variance narrative generation, replacing manual analyst effort.
- CSV / Excel Comparator — key-based reconciliation engine for UAT-versus-production validation.
- Tableau Inspector — parses Tableau data sources to map workbook lineage and impact analysis.
- Query Converter — offline Oracle-to-Starburst SQL translation utility codifying migration patterns.
- Security — SOEID authentication, session timeout, audit logging, rate limiting, CSP headers, admin panel.
- Deployment — packaged with Helm, published to GitLab, promoted through Harness pipeline onto OpenShift ECS.

IFW to Navigator Migration (Angular + TypeScript + FastAPI)

Modernized legacy workflow system from IFW to Navigator using Angular and TypeScript. Optimized performance end-to-end, reducing workflow execution time by 77% (from 2 hours to 45 minutes) while improving user experience and maintainability.

- Built responsive Angular component architecture with typed form handling, validation, and state management.
- Integrated with FastAPI backend, optimizing query performance through predicate pushdown and CTE consolidation.
- Achieved 77% performance improvement: 2 hours → 45 minutes per workflow cycle, boosting team productivity by 40%.
- Implemented real-time progress tracking and error handling with graceful degradation.

AI & Automation Engineering

Sole driver of all AI initiatives across the reporting function. Evaluate, prototype and deploy AI-assisted tooling for code generation, SQL conversion, process automation and workflow design.

- Built AI-powered process automation tooling: automated LCR commentary, SQL conversion, reconciliation engines.
- Architected automated commentary engine replacing daily manual analyst effort with variance narrative generation.
- Develop AI-assisted development workflows, establishing safe patterns for code generation within banking constraints.

Oracle → Starburst / Trino Migration Programme

Migrated and re-platformed Tableau dashboard suites across ILST, RLAP, TLST, ALCO, RLEN reporting. Resolved deep SQL-dialect incompatibilities and engineered around Trino constraints.

- Resolved critical SQL incompatibilities: Cartesian joins, BIGINT date casting, column-alias mismatches.
- Engineered around Trino's 150-stage query ceiling through CTE consolidation and UNION ALL restructuring.
- Diagnosed and escalated Parquet / Hive metastore type mismatch corrupting downstream results.

Regulatory Reporting & Domain Delivery

- Own dashboard impact analysis for major LCR reporting change expanding consolidated scope from 25 to 38 scopes.
- Mapped downstream effect of legal-entity decommission across 156 reporting scopes for dashboard replacement analysis.
- Validated new source tables against dashboard case logic to surface missing attributes before release.

Frontend Developer (React) | Nikaya

May 2023 – July 2024

- Designed and built a production React website from scratch, covering component architecture, routing, state management and dynamic content rendering.
- Integrated with backend REST APIs, handling asynchronous data flow, loading and error states.

Volunteer Web Developer | Catchafire

- Delivered website development projects for nonprofit organizations, creating custom web solutions with \$20,000 cumulative pro bono impact.